LEGEND OF TOYS · INTERNAL

Redline
Operations Manual

The self-serve guide to the production floor, QC, dispatch and admin screens that run Legend of Toys.

OPERATOR SUPERVISOR DISPATCH ADMIN

Version	Updated	Application
1.11.0	2026-06-27	redline.legendoftoys.com

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PART 1

Getting Started

What Redline is, how to sign in, find your way around, and what your role can see.

Welcome to Redline

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Redline is the production-floor cockpit for Legend of Toys. It is where the work that happens on the line (assembling, quality-checking, packing and dispatching every car and remote) turns into live numbers you can watch, plan against and act on. This manual is your self-serve guide to every screen in it.

What Redline is (and isn't)

Legend of Toys runs on three connected tools. They share one database, so an action in one shows up instantly in the others:

`Scanner`	The handheld app on the floor. This is where the physical work is **recorded**: an operator scans a unit at each station (inward, QC, packing, dispatch). Redline does not replace the scanner; it *reflects* it.
`Garage`	The store & inventory side: goods receiving, stock, the bill of materials, production-run requests and issuing parts to the floor.
`Redline`	**This tool.** The production dashboard. It plans runs, designs and sets up lines, and shows live output and quality through to packing. *Dispatch (allocation, shipments, challans) now has its own app, **Depot** —* `depot.legendoftoys.com`.

ⓘ KEY IDEA

Most Redline screens are **read-only windows** onto what the floor is doing. The numbers come from scans. If a number looks wrong, the question is almost always "what was scanned?", not "what did Redline do?". A handful of screens (Line Setup, Returns, Corrections, Repack Runs) *do* let you take action; each is called out clearly in its chapter.

The journey of a unit

Almost everything in Redline tracks a unit moving along this path. Knowing it makes every screen easier to read:

1. **Inwarded (INW):** a finished car or remote comes off assembly and is scanned in against a production run.

2. **QC Pass / QC Fail:** it is checked. Passes move on; fails go to the repair pool. *(See the QC chapter.)*
3. **Packed (PKG):** the car and its paired remote get their final sticker for a channel (Retail or Ecom).
4. **Dispatch:** the unit is allocated to a sales channel, packed into a shipment box, and shipped out. *(This now happens in the **Depot** app.)*

Returns, repairs and channel-swap repacks are detours off this main path, covered in their own chapters.

How to use this manual

Each chapter covers one Redline screen and follows the same shape: **what it is for**, **when you'd open it**, **how to read every part of it**, and **what to do** with what you see. You do not need to read cover to cover; jump to the screen you're on using the Contents page or the bookmarks panel in your PDF reader.

Role tags

Every chapter is tagged with the people who typically use that screen. If you only do one job, you can skim past chapters that aren't tagged for you.

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The next chapter, **Roles & What You Can See**, explains exactly what each tag means and which screens it unlocks.

Callout boxes

Coloured boxes flag things worth slowing down for:

> ✓ TIP
>
> A faster or smarter way to do something.

> ⚒ ON THE FLOOR
>
> Practical guidance for using a screen live, on the production-floor monitor or a phone.

> ⚠ WATCH OUT
>
> A common mistake or a number that is easy to misread.

> ⛔ IMPORTANT
>
> Something that affects real stock, shipments or money: get it right.

Signing In & Getting Around

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Getting into Redline takes one click with your work Google account. Once inside, everything is reachable from a slim sidebar organised around the production day — plan, run, check, report — plus a tucked-away Setup drawer and a search-anything command bar. This chapter shows you how to sign in and find any screen in seconds.

Signing in

Open `redline.legendoftoys.com` and choose **Sign in with Google**. Use your `@legendoftoys.com` account; only company accounts are allowed in, so a personal Gmail will be refused.

> ⚠ CAN'T GET IN?
>
> If sign-in bounces you back to the login screen, you are almost certainly signed into a personal Google account in that browser. Sign out of Google entirely, or open Redline in a fresh browser profile, then sign in with your Legend of Toys account.

The layout

Sidebar (left)	Your main menu. It groups every screen under a handful of destinations plus a Setup drawer. Click a destination with an arrow to expand its screens; only one stays open at a time. You can shrink the whole sidebar to an icon-only rail with the **Collapse** button at the bottom, and it remembers your choice next time.
Top bar	Shows where you are (the group and the screen name), an “Updated h:mm” time so you know how fresh the numbers are, and a green **Live** dot. Many screens refresh themselves automatically.
Search bar / ⌘K	At the top of the sidebar. Click it, or press `⌘K` (`Ctrl+K` on Windows) from anywhere, to jump to any screen or action by typing. This is the fastest way to get around.
Your account	Bottom of the sidebar: your name, your role, and **Sign out**.

The destinations

The sidebar follows the production day — plan it, run it, check it, report on it — rather than how the system is built:

DESTINATION	WHAT LIVES HERE
`Overview`	The day's headline view: the home screen you land on.
`Plan`	Setting up the work before it runs: Planner, New Run, Line Setup and Line Design.
`Floor`	What's happening live on the lines: Lines, Hourly, Manpower and Line Flush.
`Quality`	Checking the work: QC, Audit and Deviations.
`Reports`	Looking back: Production History and Reporting.
`Activities`	Things waiting for you: Alerts, Returns, Scans, Corrections and Repair. A red badge shows how many items are waiting (open alerts plus returns).
`Dispatch`	Dispatch now has its own app, **Depot** (`depot.legendoftoys.com`) — this group just points you there. The only dispatch-related screens still in Redline are **Repack** and **Repack Reports**.

Setup & the System Manual

Below the destinations sit two more entries:

`System Manual`	This guide, in-app and searchable, with a button to download the PDF.
`Setup`	A collapsed drawer of the do-once configuration screens: **UPC Generator**, **Operators** and **Print**. Tucked out of the daily flow, click **Setup** to open it. (Planner, Line Setup and Line Design now live under **Plan**, since you reach for them every day.)

You will only see the screens your role is allowed to open; see the next chapter.

Handy shortcuts

- **Command bar**: press `⌘K` (or `Ctrl+K`), start typing a screen or action name ("Hourly", "New run", "Print a label"), use the up and down arrows to pick, and press `Enter`. `Esc` closes it.
- **Quick search**: on any screen with a search box, press `/` to jump straight into it without reaching for the mouse.
- **Refresh**: most screens update on their own every 30 seconds. Where there is a manual `↻ Refresh` button, use it after you know something just changed on the floor.

⚒ ON THE FLOOR

Redline is built to live on the big production-floor monitor as well as a laptop. The screens you'll leave running there are **Overview**, **Lines** and **Hourly**; they auto-refresh, so they stay current without anyone touching them. Collapse the sidebar to its icon rail to give them more room. On a phone, stick to read-only screens; the action screens are easier on a larger display.

Roles & What You Can See

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Redline shows each person the screens that match their job. This keeps the floor team focused on what they need and keeps configuration screens in the hands of managers. This chapter explains the four roles used throughout this manual and which screens each one reaches.

The four roles

OPERATOR	**Operator / Floor Team.** Assembly, QC and packing staff. Mostly works from the scanner; in Redline, reads live screens to see how the day is going.
SUPERVISOR	**Supervisor / Line Lead.** Sets up lines each morning, assigns people, watches output and quality, and handles returns and repairs.
DISPATCH	**Dispatch Team.** Builds shipments, packs boxes, manages channels and challans, and runs channel-swap repacks. *The dispatch screens now live in the **Depot** app (`depot.legendoftoys.com`); in Redline this team only uses Repack.*
ADMIN	**Manager / Admin.** Full access: everything above, plus line design, operator management, UPC generation, printing, reporting and configuration.

ⓘ HOW ACCESS ACTUALLY WORKS

Your access is decided by the **role on your account**, set by an admin. The four tags in this manual are plain-language groupings of those roles; they describe *who a screen is for*, not a literal setting. If you should be able to open a screen but can't, your account role needs changing; ask an admin rather than looking for a setting on the screen itself.

Who sees what

A quick map of the screens in this manual against the roles that typically use them. A dot means "this is part of your job".

SCREEN	OP	SUP	DIS	ADM
Overview, Lines, Hourly	•	•	•	•

SCREEN	OP	SUP	DIS	ADM
QC	•	•		•
Planner, Line Setup, Manpower		•		•
Audit, Deviations		•		•
Alerts, Scans	•	•	•	•
Returns		•	•	•
Corrections		•		•
Repack Runs, Repack Reports		•	•	•
Customer Repairs, Repair Queue		•		•
Reporting, UPC, Operators, Print				•

✓ TIP

This table is a guide, not a rulebook; some people wear more than one hat (a supervisor who also dispatches, for example) and will have a role that combines access. The dots show the *primary* owner of each screen so you know who to ask about it.

PART 2

Production

Planning runs, designing and setting up lines, watching output, and quality control.

New Run / Request

`/new-run` OPERATOR SUPERVISOR ADMIN

New Run / Request is the single place production asks the store for anything. Tabs for the four run types, **Fresh**, **Outsourced**, **Repair** and **Repack**, plus an **Ad Hoc Parts** request for one-off part needs. Whatever you request lands in the store's Issue Queue, so there is one consistent "what does the store owe right now" list. (The Ad Hoc Parts tab shows only for production roles; store roles do not raise ad-hoc requests.)

WHAT IT'S FOR	READ OR ACT?	WHO	WHERE IT GOES
Requesting any run: Fresh, Outsourced, Repair, Repack.	Action.	OPERATOR SUPERVISOR ADMIN with run-request access.	The store's Issue Queue (Garage).

Production owns every run request. The store never issues stock on its own; it only fills a request that lands in its queue. Each tab below explains how the materials move, step by step, and which system handles each step.

Fresh

A normal in-house build. Pick the product, line, shift and the variant quantities (E-Comm and Retail). **You also pick the run's format.** This is required, and it tells the store exactly what to issue:

CKD	Loose parts. The store issues the full component pick list and the floor assembles the car.
SKD	Part-built bundles. The store issues the SKD bundle plus the finishing kit.
FBU	Fully built cars. The store issues the built car plus the finishing kit (battery, packaging, para); the car is issued whole, not as parts.

> ⓘ **BUILT CARS IN STOCK? THE FORM DEFAULTS TO FBU**
>
> When you pick a product that has built (FBU) units in stock, the form shows how many (for example "428 built units in stock") and defaults the format to **FBU**, so we finish those first. Switch it to CKD if you mean to build from loose parts. The format is the run's, set here by production: the store issues exactly that, or rejects the run if stock cannot match. The store no longer decides the format for you.

Outsourced

Raw build materials go to a vendor who builds the cars and returns them; the in-house team finishes them. Your part is clean and linear: **request → send → bring back**. There is no finish phase on the outsourced run itself, finishing is a separate Fresh + FBU run.

```
BUILD-MATERIALS OUT
  1. Request Outsourced Run (pick vendor)        -> store Issue Queue
  2. Store issues the BUILD materials            -> coloured car / remote parts only
  3. Hand to vendor, then Send to Vendor (Garage)   -> run goes In Progress

VENDOR builds. Built cars come back (instalments are fine).

RECEIVE  (store, in normal Receiving)
  4. Create the shipment, declare FBU, LINK this run -> built cars into normal stock
  5. Run auto-completes once the full count is received

FINISH  (in-house, a separate run)
  6. Request a Fresh run, format FBU             -> store issues the built car + finishing kit
       -> build / finish -> QC -> Packaging -> PKG Out
```

ⓘ BUILT CARS COME BACK THROUGH RECEIVING, NOT A SEPARATE BUTTON

The vendor's cars are received like any other FBU shipment: in Garage Receiving, declare **FBU** and **link this outsourced run**. They land in normal stock and the run closes itself when the count reaches the planned quantity. There is no counting pool and no Ext Inwarding scan, both are gone. Finishing is then an ordinary Fresh + FBU run by the in-house team (the built car appears as a normal part to issue).

Repair

An open-ended recovery run for the broken / returned pile. You can start it with **no target**: just open the run and the floor pulls units in as they go. Add target product lines only if you already know what you are recovering. **Pick a format** too, it classifies the repair: **FBU** = repair by swapping in a built unit, **CKD** = repair from parts. (Repair parts are still requested ad-hoc and linked to the run; the format is the classification, not a fixed pick list.)

```
  1. Request Repair Run (no target needed)
  2. Floor: at Repair Start, sticker each repairable unit and scan it in
           (scrap goes straight to the scrap pile, no scan)
  3. Repair it; when a part is short, raise an Ad-Hoc Request (Garage)
           linked to this repair run -> store Issue Queue
  4. QC -> Packaging -> PKG Out  (QC fail -> Workshop -> QC, as normal)
```

ⓘ AD-HOC PARTS ARE TRACKED PER RUN

Parts requested for a repair run are linked to that run (and product) in Garage's Ad Hoc Request screen. Over time this builds a picture of which parts repairs really need, so we can buy extra and eventually structure the run.

Repack

Swap a unit's channel packaging when you need stock in one channel but it is packed for the other. Enter the product, the **from** and **to** channel, and the quantity.

```
1. Request Repack Run (product, from -> to channel, qty)
   -> raises TWO requests:
      (a) store: issue the TO-channel packaging (box + tray)  -> Issue Queue
      (b) dispatch: release the units                         -> "to release" list
2. Dispatch: at Repack Release, scan each box to hand it over   (sign-off)
3. Repack In: scan the old box label                          (only released units)
4. Repack Out: new flipped label (channel set from the request) -> PKG Out -> dispatch
```

⚠ **UNITS MUST BE RELEASED BY DISPATCH FIRST**

This is the only place material moves backwards in the factory, so it is tightly controlled. A unit sitting with dispatch (Dispatch In or Allocated) cannot be repacked until a dispatch operator releases it at the new **Repack Release** station. Repack In hard-rejects anything that was not released. Units still on the production side (just out of PKG Out) do not need a release.

Ad Hoc Parts

A one-off parts request to the store, outside any run, for when the floor runs short. It is not a run (no units are produced); it just pulls parts. Build the request two ways:

- **BOM-based.** Pick a product and a part category; every part in that category is added, then you set quantities.
- **Manual.** Type each part code; the name fills in from the parts master.

Set line, shift and date, and optionally **link a repair run** so the parts are attributed to it (the repair feedback record). The request lands in the store's Issue Queue like everything else.

ⓘ **PRODUCTION ONLY**

Raising ad hoc parts is a production job, so the Ad Hoc Parts tab appears only for production roles. It moved here from Garage so production asks the store for everything, runs and parts, in one place.

Recent Runs

Below the request tabs is **Recent Runs**, where you track and act on the runs you have requested, grouped by stage:

GROUP	WHAT'S HERE, AND WHAT YOU CAN DO
Requested	Created but not yet issued by the store. You can **Cancel** a run before it is issued.

GROUP	WHAT'S HERE, AND WHAT YOU CAN DO
Issued	In-house runs the store has issued. **Confirm Receipt** (accept the issued quantities; a shortfall automatically raises a short-supply request for the store), and **Mark Complete** when the run is done.
Upcoming	Outsourced runs sent to the vendor, waiting for the built cars. The cars come back through the store's Receiving (declare FBU, link the run) and the run auto-completes once fully received. To turn the built cars into finished units, raise a separate **Fresh run, format FBU**.

Open ad-hoc parts requests are listed here too, with a Cancel action. Click any run to open its detail: vendor info, work orders, the pick list and the receipt status.

> ⓘ **THE STORE FULFILS FROM GARAGE**
>
> Issuing parts, rejecting a run, and sending an outsourced run to the vendor are the store's steps in Garage's Issue Queue; the built cars come back through Garage Receiving (declare FBU, link the run). The store fills your request exactly as raised or rejects it, it never changes the run. You request and track here; the store fulfils there.

Overview

`/exec` OPERATOR SUPERVISOR DISPATCH ADMIN

Overview is Redline's home screen and the factory's triage view for the day: the headline numbers, who is on the floor, and a single list of what needs your attention right now. It refreshes itself every 30 seconds, so it's built to live on the production-floor monitor all shift.

WHAT IT'S FOR	READ OR ACT?	COVERS	UPDATES
What needs me now, at a glance.	Mostly read; quick acknowledge actions.	Today (with week / month presets).	Every 30s.

The headline cards

The top row is the pulse of the day. A **Today / This Week / This Month** preset and a **New Run** button sit just above it.

CARD	MEANING
Packed	Units packed out (PKG Out flow for the period); the sub-line splits retail vs ecom.
QC Pass	Units checked-and-passed (first-pass yield).
Pass Rate	First-pass yield percentage against the 95% target (green at or above 95%, yellow below).
QC Fail	Failed QC (turns red when above zero); the sub-line splits cars vs remotes.
Repair Q	Units in the workshop (turns yellow above 50).
Pkg Out	Units packed and sitting ready-to-dispatch (at RTD).

The monthly projection chip

The **Packed** and **QC Pass** cards carry a small `↑ ~N proj / mo` line at the top. It takes what has actually happened so far this month and extends it to a whole-month estimate at today's pace: month-to-date total, divided by the day of the month, times the number of days in the month. So on the 12th, a month-to-date figure is scaled up by 30/12. Hover it for the exact basis. It is a pace estimate, not a promise, and it only sits on the two cards that count a running monthly total. The other cards are either a percentage or a live snapshot, where a monthly projection would not mean anything.

On floor today

A strip under the cards shows the day's manpower: how many people are on the floor against the plan, the present percentage and the absent count, then a per-line crew breakdown with a short marker (for example `-2`) when a line is below its planned headcount. The **Manpower** button opens the full Manpower screen.

Needs attention now

This is the heart of the screen: a prioritised list of the things worth acting on, highest pressure first. Items are worked out live from the day's data, such as a line falling behind its pace, the pass rate dropping under target, the repair queue going over cap, or open alerts, returns and audit findings piling up. Filter chips at the top (**All / High / Med / Low**) narrow the list by severity.

`Open a row`	Click any item to slide out a panel with a short trend, a plain-language recommendation, the supporting context, and a button that jumps straight to the right screen to fix it.
`Acknowledge`	Hover a row for a quick **Ack** button, or use **Acknowledge** / **Snooze 1h** in the open panel, to clear an item off your list. This only tidies your view; it does not change any floor data.

When nothing needs you, the list shows an **All clear** message.

Shift progress

On the right, **Shift progress** shows one battery per line filling toward its daily target, with a pace marker for where you should be by now (green when on pace, red when behind).

Tomorrow's runs

At the bottom, a table of the production runs already lined up for tomorrow: those requested, submitted or in-flight (closed and cancelled runs are left out). Columns are Run, Product, Line, Target (with the retail / ecom split) and Status. Click a row to open New Run. Use it at end of shift to see what the floor is set up to build next.

> ⚠ ABOUT THE DATE PRESETS
>
> **Today** shows today's figures. **This Week** and **This Month** re-key the headline cards to running totals for that range; where a range figure is not available a card shows a dash. The monthly projection chip always reads the month so far, whichever preset is selected.

⚒ **ON THE FLOOR**

This is the screen to leave running on the wall. Pace and behind/ahead only appear once the shift is underway and a target is set, so an empty pace early in the morning is normal.

Planner

`/planner` SUPERVISOR ADMIN

The Planner turns the month's dispatch demand into scheduled production runs. You upload the demand plan, the Planner shows what's needed by which date and how much current stock covers it, and you build and create the runs to fill the gaps, without leaving the page.

WHAT IT'S FOR	READ OR ACT?	WHO	STARTS WITH
Scheduling production to meet dispatch demand.	Action: creates real runs.	SUPERVISOR ADMIN	A demand-plan CSV upload.

Step 1: Upload the demand plan

Use ↑ **Upload Plan CSV** at the top right. The file must be the **Projection V2 tab** of the Dispatch Plan Google Sheet, exported via *File → Download → CSV*.

> ⛔ **USE THE RIGHT TAB**
>
> Exporting the wrong sheet tab is the most common error; the upload will reject it with a message about a missing SKU, Channel or Mapping column. Rows with a dispatch date in the past are silently dropped, and an all-zero or unrecognised-mapping file shows "No demand rows found". Always export the **Projection V2** tab.

The three tabs

Dispatch Timeline

What's needed, by date. Each dispatch date is an expandable card showing total units, product count, and `⚠ N gap` in red where stock falls short. A toggle switches between **Cascaded Waterfall** (stock carries forward across dates per product) and **Normal** (each date against current stock). Expanding a product shows its variants with columns Need, RTD, Allocated, Balance/Available, Gap and a status pill.

Recommended Runs

The working tab. On the **left**, dispatch needs by date, each with a suggested production date (or `⚠ Too late to produce in time`) and a **+ Schedule** button per product. On the **right**, your **Production Schedule**: the "carts" you're building.

1. **Click + Schedule on a product.** A panel opens with editable Qty Ecomm / Qty Retail per variant (each showing how much gap remains), a production date, and an L1/L2/L3 line.

2. **Add to Schedule.** The line drops into a cart on the right. Build up as many lines and days as you need (**+ New Day** starts a fresh cart).
3. **Adjust in the cart.** Edit quantities inline; `↻` repeats a line to another day, `×` removes it. A yellow `⚠ Stock short` warning flags any line whose parts are insufficient.
4. **Create All for the date.** One button creates a real production run for every line in that cart with a quantity above zero. Each line reports `✓` + run number, or an error.

Batch Sizes

Set the default run size per product (with the most common past size shown alongside). This drives how many runs the Planner suggests to cover a gap. Edit inline and Save.

Reading the status pills

PILL	MEANING
Covered	No gap: stock meets the need.
Plan Run	A gap exists; schedule a run with comfortable lead time.
Urgent	Tight on time: schedule today.
Critical / Too Late	Today-only, or already past the point of producing in time.

> ⚠ WATCH OUT
>
> A product can't be scheduled on the same line and date twice; the Planner blocks it. Creating runs sends one request per line, so a large cart takes a few seconds; if a duplicate/open-run already exists you'll be asked whether to create another anyway or skip it.

Line Design

`/line-design` ADMIN

Line Design defines the physical layout of each product's assembly line: how many work stations sit in each department and how many people each holds. It's the template that Line Setup staffs every morning. Designs are versioned, so layout changes are tracked over time.

WHAT IT'S FOR	READ OR ACT?	WHO	FEEDS
Designing the station layout per product.	Action: create/edit/version designs.	ADMIN / floor supervisors.	Line Setup's staffing.

The screen

On the **left**, the list of products that have a design, each with its active version (e.g. `v3`) and effective date. On the **right**, the editor for the selected product: a summary bar of worker totals per department, then the four department sections each holding station cards.

Departments are always **Prep → Assembly → QC → Packaging**, and stations are auto-coded `PR-01`, `AS-02`, `QC-01`, `PK-03` by department and position.

Working with station cards

`Capacity`	Click the person icon to toggle a station between one worker (👤) and two (👤👤).
`Car / Remote`	Prep and Assembly stations can be tagged **CAR** (blue) or **REM** (purple) to mark which stream they handle. Click to tag, click again to clear.
`Reorder / remove`	Drag a card to reorder within its department; `×` removes it; the "+ Add station" tile adds one.

Creating and versioning

1. **+ NEW** creates the first design for a product. Pick the product, an effective-from date, optionally copy stations from another product's design, and add notes.
2. **Edit, then SAVE CHANGES.** The save button stays greyed out until you actually change something; nothing is saved until you click it.

3. **NEW VERSION** when the layout genuinely changes. It carries the current stations forward as a starting point and requires a later effective date and a change note. The old version is kept in **Version History**, read-only.

> ⓘ ONE DESIGN PER PRODUCT
>
> A product can have only one line design. Once created it disappears from the New-design list; change it with **SAVE CHANGES** for small edits, or **NEW VERSION** when you want to preserve the old layout in history. To start a similar product, use the “copy stations from” option.

Line Setup

`/line-setup` SUPERVISOR ADMIN

Line Setup is the morning staffing screen. For a chosen date it takes the operators who have clocked in and lets you place them on the physical stations of each line, guided by who's worked there before. When you're done, print a roster for the floor.

WHAT IT'S FOR	READ OR ACT?	WHO	NEEDS FIRST
Assigning clocked-in staff to stations.	Action: assign / unassign / print.	SUPERVISOR ADMIN	Attendance done; a line design.

The layout

Three columns (**L1, L2, L3**) side by side. Each shows the run scheduled on that line (or `NO RUN`), department chips reading `staffed/total` (green when full), and the station cards from that line's design with one slot per worker.

Assigning people

1. **+ Assign on an empty slot.** A picker opens with a search box and a suggestion list. Suggestions show frequency dots (`●` worked here before, `●●` 3+ times, `●●●` 6+ times), so you can place experienced hands first.
2. **Pick a name.** They're assigned immediately, no confirmation. The `×` on a filled slot unassigns them, also immediately.
3. **Suggest All** auto-fills every open slot in a department with the best-matched available people in one go.
4. **Print Roster** opens a printable sheet for that line (enabled once at least one person is assigned).

> ⚠ **CLOCK IN FIRST**
>
> Only **clocked-in** operators appear as suggestions. If nobody is clocked in, you'll see "No clocked-in operators available". Run attendance in **Manpower** first. If a line has a run but no line design, you'll be prompted to create one in **Line Design**.

ⓘ **NO UNDO PROMPT**

Assign and unassign take effect instantly with no confirmation; a mis-click is fixed by simply re-assigning. An operator can only be on one line per day; placing them somewhere new moves them.

Lines (Live View)

`/lines` OPERATOR SUPERVISOR DISPATCH ADMIN

Lines is the live floor monitor: each production line's real-time output today, the people on it, and how fast each station is moving. Like Overview, it's built to run on the wall and refreshes every 30 seconds.

WHAT IT'S FOR	READ OR ACT?	COVERS	UPDATES
Watching each line's live throughput.	Read-only.	Today only, lines L1–L3.	Every 30s.

Line Performance

One card per line, each showing the product on it, a big colour-coded **% complete** against the run target, and a five-stat grid:

STAT	MEANING
INW	Units inwarded (sub-line shows remotes).
QC Pass (green)	Passed QC; car/remote split below.
QC Fail (red)	Failed QC; car/remote split below.
PKG (yellow)	Units packaged.
Dispatched (green)	Retail + ecom shipped (sub-line splits R · E).

The footer adds operator count, downtime minutes, percent done, target, first-pass yield (**FPY**) and the first-scan time. The big % is green at 90%+, yellow at 60%+, orange at 30%+, red below.

Operator Output

A per-operator table: INW cars/remotes, QC pass cars/remotes, QC fails, and pass rate (green 95%+, yellow 85%+, red below). This is how you see who is moving and who may be struggling.

Station Pace (Takt)

Average time per unit at each station (INW, QC, PKG, and the shared PKG_OUT) across the lines. **Faster is greener**: green up to 5 min, yellow up to 10, red beyond. A slow, red station is your bottleneck.

⚠ TAKT REFRESHES ONCE

The line stats and operator output refresh every 30 seconds, but Station Pace is measured only when the page *opens*, so it can look stale next to the live numbers. Reload the page to refresh the pace figures.

Manpower

`/manpower` SUPERVISOR ADMIN

Manpower is the people screen: who is on the floor right now, daily attendance, the day's roster, performance points, and the shift timings behind it all. It's organised into tabs — a live floor view, **Attendance** (production), a separate **Dispatch** attendance view, the Daily Roster, Performance, Manpower analytics, and **Shifts**. (Store attendance and store shifts live in Garage.)

WHAT IT'S FOR	READ OR ACT?	WHO	UPDATES
Staffing the floor and tracking attendance.	Live View reads; the rest act.	SUPERVISOR ADMIN	Live View every 60s.

Live View

Today's headcount across L1, L2, L3, Dispatch, Store, Others and Unassigned, then operator cards under each line. It shows only people with an **open shift** (clocked in, not yet out); someone who has clocked out drops off. "On floor" (from attendance) and "assigned" (from the roster) are different counts, shown side by side. Refreshes every 60 seconds.

Attendance

Clock-in/out records for a chosen date, for the **Production** team (assembly, QC, packaging, admin). Dispatch has its own tab and Store attendance is in Garage. Columns: Clock In (with a `+Xm late` note when someone arrived after their shift start), Clock Out (with `+Xm OT` when they stayed past the end), Duration, **Streak**, **Absent (mo)**, **Day status** and Device. A shift with no clock-out shows an amber `Open ⚠` badge and gets a **Close Shift** button. Use the **Department** filter at the top to narrow the list.

- **Streak** — how many working days in a row the operator has been present, counting back from the selected date. Working days are Mon–Sat; Sundays neither count nor break a streak.
- **Absent (mo)** — how many working days (Mon–Sat) the operator has missed so far this month. Days before they joined don't count.
- **Day status** — a manual classification you set per row: `Normal` / `Full day` / `Half day` / `Absent` / `Leave` / `Holiday`. Use it when someone was sent home, took a half-day or was on leave; it feeds the future payroll/overtime engine. Leave it on Normal for an ordinary worked day.

> ⓘ AUTO CLOCK-OUT OVERNIGHT
>
> Anyone still clocked in from a previous day is clocked out automatically overnight and tagged `Auto`. With shifts live, the clock-out is stamped at that person's **scheduled shift end** (older records that predate shifts fall back to 1:00 AM IST). This credits a forgotten clock-out to the end of the shift they actually worked — you don't need to Close them by hand.

> ⚠ CLOSE SHIFT SETS CLOCK-OUT TO NOW
>
> Closing an open shift stamps the clock-out at the current time and can't be undone here. Use it to tidy up someone who forgot to clock out *today*, or to close out someone who genuinely left early (they can't scan out before their shift ends — see Shifts). The duration is calculated from it.

Dispatch

The same attendance view, filtered to the **Dispatch** team. Dispatch runs two shifts that overlap in the middle of the day, so each dispatcher is given a **home shift** (set on the Shifts tab) that decides which shift a scan belongs to. Everything else matches the Attendance tab — clock times, late/OT, streak, day status and Close Shift.

Daily Roster

Assign clocked-in people to lines and stations for the day. Set target headcounts per line/station at the top (pre-filled from the active run's line design), then staff from the **Available** pool: the people who are clocked in but not yet assigned.

- **Drag** a chip (or several selected chips) onto a station section, or use the `+` picker.
- **Auto-assign** fills the target slots in order from the available pool, then reports how many were placed and how many slots were left open.
- `×` removes someone from a line.

> ⓘ ATTENDANCE COMES FIRST
>
> "Available" only lists people who are *clocked in*. If someone isn't showing, they haven't clocked in yet. Sort attendance before rostering.

Performance

Pick an operator to see their points total (green if positive, red if negative) and event history. **+ Add Event** logs a point change: choose points (positive for good, negative for poor), a category (Attendance, Quality, Behaviour, Output, Other), a date and a reason. Every event is stamped with who recorded it.

Manpower analytics

A read-only view that turns attendance into a planning tool: who comes regularly, who's irregular, and how it's trending — so you can keep your reliable people on the critical line stations and move irregular ones to lower-risk work. Pick a window at the top (**30 / 60 / 90 days** or **this month**; 60 is the default). Operators are grouped into **department sections**, with the departments carrying the most absenteeism shown first, and the least-reliable people at the top of each section.

Each operator carries a **reliability tier** based on how many of the working days (Mon–Sat) in the window they were present:

- **Reliable** — 90% or more. Your continuity backbone; best kept fixed on a line so on-the-job training compounds.
- **Steady** — 75–90%.
- **Irregular** — 50–75%. Candidates to move to low-risk activities.
- **Critical** — under 50%.

The **heartbeat strip** is the quick read: one square per working day, newest on the right — `green` = present, `red` = absent, grey = before they joined. A clean green run is a dependable operator; scattered red is sporadic; a repeating gap (e.g. every Saturday) is a predictable pattern you can plan around — flagged as an "Often off <day>" hint. Each row also shows the current / longest streak, a trend arrow (recent half of the window vs the earlier half), and average arrival time with a count of days the person arrived later than their own norm.

> ⓘ HIDE NEVER-PRESENT
>
> By default, operators with zero attendance in the window are hidden (usually departed or test accounts) so the genuinely irregular people stand out. Untick **Hide never-present** to show them.

Shifts

Each team sets its own shift timings here — the start and end that drive the clock-in window, lateness and overtime on the scanner. **Production and Dispatch** are managed on this tab; **Store** manages its shift in Garage. Shift names are free text.

- **Edit timing** writes a *new version* effective from the date you pick — it never overwrites the old one. The change takes effect that day; previous versions stay on record under **History** (so you can always see what the timing was on any past date, and who changed it).
- **Add shift** creates another shift for a department (e.g. Dispatch's second shift). **Disable / Enable** turns a shift off without deleting it.
- **Dispatch — assign operators**: the two dispatch shifts overlap, so a scan time alone can't tell them apart. Set each dispatcher's **home shift** here so their clock events are attributed to the right one.

⚠ NO EARLY CHECKOUT

The clock-out window opens only **at the shift's end time**. People can clock in early and clock out late, but they cannot check out early — a too-early scan just says "Already Clocked In". For a genuine early departure, close the shift here or set a **Half day** day-status; the overnight auto-close otherwise credits them to the shift end. Fine-tuning (how early clock-in opens, the overtime grace, the minimum time on site) is under **Advanced** in Edit timing.

Hourly

`/hourly` OPERATOR SUPERVISOR DISPATCH ADMIN

Hourly is a heatmap of output through the day: how many units cleared Inward, QC and Packaging in each hour, per line. Open it to see whether the floor is pacing steadily or stalling, and exactly which hour things slowed.

WHAT IT'S FOR	READ OR ACT?	COVERS	UPDATES
Watching throughput build hour by hour.	Read-only.	The selected date, lines L1–L3.	Every 30s.

How it's laid out

Three stacked panels (**Inward**, **QC Pass** and **Packaging**) each showing the day's grand total on the right. Inside each, one row per active line (L1 yellow, L2 blue, L3 green) with a strip of hour cells from 9 AM through 6 PM (later hours get a `+` for overtime). Each cell is a little bar that fills against that line's busiest hour, with the count printed inside.

⚠ EMPTY CELLS AREN'T ZERO

Hours that haven't happened yet render faint and empty; that means "not reached", not "zero produced". Only past hours with no bar are genuine zeros.

⚒ ON THE FLOOR

Read it top to bottom: if Inward is flowing but QC's hourly bars are short, units are stacking up at the QC bench. A line that goes quiet for an hour mid-shift is the signal to check what happened: a parts shortage, a breakdown, or a break that ran long.

QC

/qc OPERATOR SUPERVISOR ADMIN

The QC screen is your daily quality scoreboard. It does not pass or fail units (that happens on the scanner during the quality check), but it turns every one of those pass/fail scans into a clear picture of **how fast QC is moving**, **how much is passing first time**, and **what is going wrong and where**. On today it refreshes itself every 30 seconds, so you can leave it open and trust the numbers; pick an earlier date and it stops refreshing, because a past day is settled and the figures should not move under you.

WHAT IT'S FOR	READ OR ACT?	COVERS	UPDATES
Spotting quality problems on the floor as they happen, today.	Read-only: a live view.	Any single day (opens on today), all lines.	Every 30s on today; a past day is fixed.

ⓘ **WHERE THE NUMBERS COME FROM**

When a unit is inwarded it gets an **INW** scan; when it's checked it gets a **QC Pass** or **QC Fail** scan, and a failure records one or more **defect codes**. This screen simply adds those scans up for today. Nothing here changes a unit; to actually pass, fail or re-check a unit, use the scanner on the floor.

QC Cycle Time: INW to outcome

This is how long units are spending in quality control: the time from a unit being inwarded (INW) to it getting its QC Pass or Fail. Shorter is better; long cycle times mean units are piling up at the QC bench.

The summary cards

CARD	WHAT IT TELLS YOU
Avg Cycle Time	The average across all lines, with the number of units it's based on. **Colour-coded:** green up to 30 min, yellow up to 60 min, red beyond. This is your headline number.
Median	The middle unit's time. If the median is much lower than the average, a few slow units are dragging the average up.
Avg, QC Pass	Average time for units that passed (green). Usually the fastest.
Avg, QC Fail	Average time for units that failed (red). Often longer, as failures take more inspection.

CARD	WHAT IT TELLS YOU
Fastest	The quickest unit through QC today.
Slowest (Normal)	The slowest unit that still counts as normal; genuine outliers are excluded (see below).

⚠ THE OUTLIER STRIP

If you see a red strip saying *"N outlier units excluded"*, that's deliberate. A unit that sat overnight or was set aside for hours would wildly distort the averages, so the screen fences those off and tells you how many it removed and what the longest one was. The averages above the strip are the *true* working pace; the strip is just honesty about what was set aside.

Per-line breakdown

Below the summary, each active line (L1, L2, L3) gets its own card with the same figures. Use this to compare lines; if one line's average is red while the others are green, that's where to look first.

First Pass Yield by line

First Pass Yield (FPY) is the single most important quality number: **the percentage of units that passed QC the first time**, without needing repair. Each row is a line + product + date, with a coloured bar and a percentage.

⚠ **"Fail" means failed — it is not the same as "did not pass first time".** The headline now reads *first-pass · rework · fail*, three separate numbers. **Rework** is a unit that failed earlier, was repaired and then *passed* on a re-check — it is a pass, not a failure. **Fail** counts only units with a QC Fail scan on the day. Until 22 Aug 2026 the screen added the two together and called the total "fail", which on one morning showed 61 failures on a day with none at all. The percentage did not change and was never wrong: a unit that needed a second go genuinely is not a first-pass, so rework still pulls FPY down. That is the point of the measure — rework is waste even when the unit ends up passing.

Pick a date. The date sits at the top left. Use the arrows to step a day either way, or click the date for a calendar; a **Today** button appears once you are on a past day. You cannot pick a future date.

COLOUR	FPY	WHAT IT MEANS
Green	95%+	Healthy. The line is building right the first time.
Yellow	85–94%	Watch it. Something is slipping; check the defects below.
Red	under 85%	Act. Too many units failing first time; find the cause now.

✓ TIP

A red FPY plus a high QC-Fail cycle time on the same line is a strong signal: that line is both failing more *and* taking longer to do it. Pair the two sections together when you diagnose.

Top Defects

For each line, the most common defects today, split into two columns:

- **Functional** (red total): the toy doesn't work right: motor, electronics, movement.
- **Visual** (orange total): how it looks: scratches, paint, stickers, finish.

Each defect appears as a small card showing its **code**, a short **description**, its **category**, a **severity** (Critical, Major, Minor or Cosmetic), and how many times it occurred. Cards are ordered worst-first: Critical before Major before Minor before Cosmetic, then by count. A FLAG marker means the defect is flagged for operator training.

ⓘ COSMETIC DEFECTS ONLY START APPEARING FROM 28 AUG 2026

The scanner has always offered the five **Cosmetic** defects (body scratch or scuff, sticker issue, foreign material), but a database rule silently rejected them, so the QC Fail scan failed and nothing was recorded. That was fixed on 28 Aug 2026. Expect Cosmetic counts to appear from that date onward — **a jump in cosmetic defects is the capture starting to work, not quality getting worse**, and comparisons with earlier months will understate them.

Defect Breakdown by product

The same defects, organised by **product** instead of line. Each product is a row showing its total defect count; **click a row to expand it**. Inside, defects are grouped first by component (Car vs Remote), then by severity, then listed individually with their code, description and count. A TRAINING tag again marks defects flagged for coaching.

⚒ ON THE FLOOR

Use *Top Defects* when you're asking "which line needs help right now?" and *Defect Breakdown* when you're asking "which product is giving us trouble?". Same data, two lenses.

Repeat Failures

The last section is a table of individual units that failed with **more than one defect**: the units giving the most trouble. Each row shows the unit's UPC, product, total defect count (red), how many unique defect codes it hit, and the list of codes.

> **IMPORTANT**
>
> A unit appearing here with many defects is a candidate for a deeper look before it's re-worked; repeated, varied failures on one unit can point to a damaged part or a build that should be pulled rather than patched. Cross-check its UPC in the **Scans** screen to see its full history.

A typical QC check-in

1. **Glance at the Avg Cycle Time card.** Green and steady? QC is keeping pace. Red? Units are backing up; check which line in the per-line cards.
2. **Scan the FPY bars.** Any red or yellow line is your priority.
3. **For a problem line, read its Top Defects.** Functional or visual? Critical or minor? That tells you whether it's a parts/assembly issue or a finishing issue.
4. **If a product keeps recurring, expand it in Defect Breakdown** to see exactly which defects and components are driving it.
5. **Check Repeat Failures** for any single units that need pulling rather than re-working.

QC Audit

`/audit` SUPERVISOR ADMIN

QC Audit is where supervisors run walk-the-floor quality rounds, record what they find, and chase every finding through to fixed. It has two tabs: **Log** for running rounds today, and **Tracker** for managing open findings over time.

WHAT IT'S FOR	READ OR ACT?	WHO	SEVERITY
Running QC rounds and resolving findings.	Action: rounds and findings.	SUPERVISOR ADMIN	Critical · High · Medium · Low.

Log: running a round

A day summary strip (rounds today, critical/high open, total findings, confirmed) sits above the round cards. To run a round:

1. **+ Start New Round.** A round opens, marked Active (green).
2. **+ Add Finding** for each issue. Pick a **Line**, a **Category** (e.g. QC Pass: Damage, Assembly: Wrong/Missing Part, SOP Deviation), a **Severity**, a description, and the action required.
3. **Close Round** when the walk is done. Closed rounds still expand, and their findings can still be resolved and confirmed.

Tracker: closing the loop

Filter all findings by line, status, severity, category and date. Per-line scorecards show open counts and the oldest open item; a card border turns red when its oldest open finding is **overdue**. The findings table carries the action buttons.

STATUS	MEANING & NEXT STEP
Open	Raised, not yet fixed. A supervisor marks it **Resolved** with a required note describing the fix.
Resolved	Fixed, awaiting sign-off. The round's auditor (or a super admin) clicks ✓ **Confirm Fixed**, or anyone can **Reopen** it.
Closed	Confirmed fixed, shown dimmed. Done, not gone.

A **Repeat Offenders** panel flags findings that have recurred in the last 30 days: the patterns worth fixing at the root.

SEPARATION OF DUTIES

The supervisor who *ran* a round cannot mark its own findings resolved; a different supervisor must. The round's auditor (or a super admin) then confirms the fix. Resolving always requires a written note of what was done; blank notes are rejected.

OVERDUE RULES

An open Critical or High finding is overdue after 24 hours; Medium or Low after 48. Overdue ages show red and turn the scorecard border red: your prompt to escalate.

Line Flush

`/line-flush` OPERATOR SUPERVISOR ADMIN

A line flush is how production returns leftover material to the store at the end of a run: the unused, damaged, QC-rejected or partly-assembled parts that did not go into a unit. You raise the flush here; the store receives, dispositions and verifies it in Garage.

WHAT IT'S FOR	READ OR ACT?	WHO	THEN
Returning leftover material to the store.	Action: raises a flush.	OPERATOR SUPERVISOR ADMIN with flush access.	Store verifies it in Garage.

Raising a flush

1. **Click New Flush.** Set the date, line, shift, and pick the flush type: against a **Production Run** or **Standalone**.
2. **Add the parts being returned.** For a run flush, use **Load From Run** to pull the run's pick list, or add parts by code. Each part can be split across return types (Unused, Damaged, QC Rejected, Partial Assembly, Wrong Issue) with a quantity per split.
3. **Submit Flush.** It is sent to the store with a flush id, marked Pending Verification.

Tracking flushes

The list shows every flush with its status: **Pending Verification** (waiting for the store), **Verified** (the store counted and accepted it) or **Disputed**. Open one to see the parts you returned and, once verified, the store's counted quantities, any variance, and the dispositions they applied.

> ⓘ **VERIFICATION HAPPENS IN GARAGE**
>
> Raising a flush is production's job and lives here. Counting the returned material, setting dispositions (back to stock, quarantine, scrap, rework) and accepting it into inventory is the store's job, on Garage's **Flush Verify** screen. The store's physical count is the figure that actually moves stock, so the detail you see here is read-only.

Process Deviations

`/process-deviations` SUPERVISOR ADMIN

Process Deviations is the full record of process changes, temporary or permanent, and their supervisor review. It is where someone documents a change to how a step is done, routes it for the right level of sign-off, and acts on it through its whole life: propose, approve or reject, acknowledge, and close. This is purely a production tool; the store has no process deviations.

WHAT IT'S FOR	READ OR ACT?	WHO	TIERS
Documenting & approving process changes.	Full management: propose, approve, close.	OPERATOR proposes; SUPERVISOR ADMIN approve.	L1 / L2 / L3 by severity.

The status tabs

Tabs across the top filter the list: **Pending Approval**, **Active on Floor**, **Retro Sign-off**, **Rejected**, **Closed**, and **All**. The **Active on Floor** tab is laid out as cards grouped by line, so anyone can see at a glance what is in force on each line right now: each card shows the deviation number, severity, title, station or type, and when it expires or that it is open-ended. **Operators must follow active deviations.** Every other tab is a table you can click into.

Working a deviation

Click any row to open its full detail: the description, reason and rollback plan, line/run/product/station, the approval window, and a complete history of every action taken on it. From here, depending on your permission and the deviation's current tier, you can:

- **Approve** or **Reject** (reject needs a reason) when it is pending at your tier.
- **Escalate** a pending deviation to the next tier.
- **Acknowledge** an approved, rejected or closed deviation, so there is a record you have seen it.
- **Confirm or reject a retroactive sign-off** for a reactive deviation applied while you were out.
- **Close** an approved deviation (reason required) once it no longer applies.

Proposing a deviation

+ Propose Deviation opens a form: pick a **type** (Material Substitution, Step Skip, Quality Criteria Relaxed, Process Workaround…), a **severity**, the line and station, a title and description, an optional reason and rollback plan, and an optional “effective from / until” window. The selected severity shows its approval rule:

SEVERITY	WHO APPROVES
Low	L1 supervisor.
Medium	L1 + an L2 second pair of eyes.
High	L3 admin only.
Critical	L3 admin only; reactive auto-approval blocked.

Tick ⚡ **Reactive** if the change is already in effect and you are logging it after the fact. Low and medium reactive deviations are created as approved immediately but flagged for a supervisor's retro sign-off; High and Critical still wait for L3 sign-off regardless.

> ⓘ **THE FULL RECORD LIVES HERE**
>
> This screen is the complete deviations queue and history, moved into Redline so production owns it end to end. The permissions that decide who can approve at each tier are still set in Garage's Users screen (the company-wide role matrix); this screen simply reads them.

Repack Runs

`/repack-runs` DISPATCH SUPERVISOR ADMIN

Repack Runs track “channel swaps”: when already-packed units need re-doing for a different sales channel (say retail → ecom). A run sets a target number of units to repack; operators then physically scan boxes at the Repack In and Repack Out stations, and progress accrues against the run. This screen opens, tracks and closes those runs; the actual repacking happens on the scanner.

WHAT IT’S FOR	READ OR ACT?	WHO	REPACKING
Tracking channel-swap repacks.	Create / complete / cancel runs.	DISPATCH SUPERVISOR ADMIN	Done at the scanner stations.

How a channel swap works

Request the run from **New Run / Request → Repack** (product, from → to channel, quantity). That raises two requests automatically: the store issues the **to-channel packaging** (box and tray) into the Issue Queue, and dispatch gets a **release list**. Then the physical work happens on the floor:

1. **Dispatch releases the units at Repack Release.** A dispatch operator scans each box to hand it over to the run. Units still on the production side (just out of PKG Out) skip this.
2. **Scan the old box at Repack In.** Only released (or PKG-Out) units are accepted; the unit rolls back to QC Pass and everything downstream is wiped, ready to re-box for the new channel.
3. **Re-pair at Repack Out.** The operator re-confirms the car + remote pair; the new channel comes from the request and a fresh sticker prints.
4. **The unit rejoins dispatch.** From there it flows through the normal pipeline (PKG Out → allocate → pack → dispatch) unchanged. The old packaging goes back to the store via line flush.

Tracking it from this screen

This screen tracks the runs requested from New Run / Request. The list shows each run’s progress (`repacked / target` + %) and status: **Open**, **In Progress**, **Completed** (green), **Cancelled**. A run’s detail page tracks each swap as it happens: car UPC, paired remote, the `from → to` channel, old and new labels, and in/out times. Units scanned in but not yet re-packed out show as **in-flight**. **Complete** closes the run; **Cancel** stops it (units already repacked stay repacked).

ⓘ **REQUESTS COME FROM NEW RUN / REQUEST**

Repack runs are now requested with a product, channel direction and quantity (not just a bare count), which is what lets the store auto-receive the right packaging and dispatch know which units to release. This screen is where you watch and close them.

PART 3

Inbox

The one activity surface: live alerts, returns intake, the raw scan log, and fixing mistaken scans. In the app these sit under the Inbox destination alongside the Repair queue.

Alerts

`/alerts` OPERATOR SUPERVISOR DISPATCH ADMIN

Alerts is the floor's exception list: scans that were rejected or flagged by the rules, such as an over-target pack, a wrong-station scan, or a failed check. Open it to review what went wrong over a date range and mark each one as handled.

WHAT IT'S FOR	READ OR ACT?	WHO	BADGE
Reviewing and clearing scan violations.	Reads; one action: acknowledge.	SUPERVISOR ADMIN (+ all can view)	Red count in the sidebar.

Reading it

Four overview cards head the screen: **Total**, **Unacknowledged** (red when above zero), **Worst Line** and **Worst Station**. The table below lists each violation with its time, line, station (colour-coded), operator, the unit's UPC, and the violation message in red. The status badge reads `● Open` (red, needs attention) or `✓ Ack` (green, handled); acknowledged rows are dimmed.

Filter by status (All / Unacknowledged / Acknowledged), by line, and by station, and set the From/To dates (the **Today** chip resets to today).

Acknowledging a violation

1. **Click Ack on an open row.** A box shows the violation detail and an optional note field.
2. **Add a note (optional) and Confirm Ack.** The row turns green and dims; it's marked as seen and dealt with.

> ⓘ **ACKNOWLEDGING ISN'T FIXING**
>
> "Ack" records that a supervisor has seen and handled the alert; it doesn't change the scan or the unit. The aim each day is to drive **Unacknowledged** to zero, having actually dealt with the cause on the floor.

Returns

`/returns` SUPERVISOR DISPATCH ADMIN

Returns is a floor view of the return piles (**UDR, CXR, BRV, Loss and Scrap**) grouped by variant, with KPI totals. From here Production can **request a repair run**. Dispositioning and issuing happen in Garage.

WHAT IT'S FOR	READ OR ACT?	WHO	UPDATES
Watching piles + requesting repair runs.	Mostly read; request a run.	SUPERVISOR DISPATCH ADMIN	Every 60s.

The piles

`UDR`	(green) Undamaged: re-dispatch as-is via PKG OUT, no repair.
`CXR / BRV`	(yellow / blue) Customer and bulk/vendor returns: go to a repair run.
`Loss`	(red) Switcheroo / wrong / not received: written off.
`Scrap`	(grey) Units scrapped on the repair line.

Each pile is grouped by product / model / colour with a count (and oldest age for UDR/CXR/BRV). The KPI cards sum each pile.

Request a repair run

Request Repair Run opens a picker of the CXR/BRV buckets and a line. It creates an empty `REP-NNN` run with those target lines; the **Store** then physically issues (scans out) the units against it on Garage → Issue Repair.

> ⓘ **PAIRING IS RE-MADE AT QC PASS**
>
> Returned cars and remotes move through repair independently; a fresh car↔remote pair is created at the QC PASS station, like new production. Dispositioning and issuing are done in Garage.

Scans

`/scans` OPERATOR SUPERVISOR DISPATCH ADMIN

Scans is the master log of every barcode scan on the floor (inwarding, QC, workshop, packing, returns and more) with daily totals. Open it to trace a single unit's full history, or to review a day, week or month of activity.

WHAT IT'S FOR	READ OR ACT?	COVERS	SHORTCUT
Looking up scan history and totals.	Read-only.	Any date or range; any UPC.	`/` focuses the UPC search.

Two ways to use it

1. **Review a period.** Pick a date or a Today / This Week / This Month preset. The summary cards count each activity for the day (INW and QC Pass also split cars vs remotes); click any card to filter the table to that activity. **Load More** pages through the list.
2. **Trace a unit, or find a product.** Type into the search box. **3 or more characters** searches all history across every date; 1–2 characters just filters the rows already on screen. As well as a UPC or box label, you can search by **product, variant or colour** — it matches the same text the PRODUCT column shows, so pasting what is on screen finds it.

Reading the table

Columns: Time, UPC, **Activity** (colour-coded badge), Line, Product, Operator, **Loop** (how many times the unit has cycled through), and Status: `✓ OK` or `✗ Voided`. Tick **Show Voided** to include cancelled scans (they show dimmed). Activity badges follow the floor's colour key: INW blue, QC Pass green, QC Fail red, WKS purple, PKG violet, RTO/RTD orange-teal.

⚠ TWO THINGS THAT TRIP PEOPLE UP

A **short** search (1–2 characters) only filters the current day's loaded rows; it won't find scans on other days. Type 3 or more characters for a true all-history lookup. **A search returns at most 500 matches** — a product search often has more than that, so when it is capped the screen says so underneath the table. Narrow the search to see the rest. Also, the summary cards always count the *start* date, even when a multi-day range is selected for the table.

Corrections

`/corrections` SUPERVISOR ADMIN

Corrections is the controlled way to fix a mistaken scan. Supervisors can **void** a wrong scan; managers can **amend** a scan's line or notes. Every change is logged and kept; nothing is silently deleted.

WHAT IT'S FOR	READ OR ACT?	WHO	AUDIT
Fixing scans recorded in error.	Action: void & amend.	SUPERVISOR (void) ADMIN (amend)	Every change is logged.

The two tiers

ACTION	WHO	WHAT IT DOES
Tier 2: Void	Supervisor	Cancels a wrong scan made in the *current shift*. The scan stays in the log marked Voided. A reason is required.
Tier 3: Amend	Manager	Corrects a scan's line or notes on any scan, leaving an immutable audit trail. A reason is required.

Doing a correction

Find the scan (pick the date, optionally filter by activity or search a UPC). Then use the action button on its row:

1. **Void.** Confirm the scan shown, type a required **reason**, and Confirm Void. The row becomes Voided.
2. **Amend.** Change the **Line** and/or **Notes**, type a required **reason** (the Confirm button stays disabled until you do), and Confirm Amend. It rejects "no changes to save" if nothing was actually changed.

> ⚠ GOOD TO KNOW
>
> You'll only see the action buttons your permission allows: void needs supervisor rights, amend needs manager rights, and an already-voided row offers neither. Void is meant for the current shift; an older scan may be refused by the system even if the button shows. Amend here changes line and notes only.

PART 4

Scanner

The production stations on the floor scanner: how to get in, and what each station does. (Dispatch stations now live in the Depot manual.)

The Floor Scanner

OPERATOR SUPERVISOR DISPATCH ADMIN

The floor scanner is the handheld PWA your team uses at every physical station. The whole production line and the whole dispatch line run through it. This chapter covers how the scanner is now organised: how you get into your department, sign in, and reach the station you need. The station chapters that follow cover each scan in detail.

WHERE	YOUR AREAS	WHO	GETS YOU IN
`scanner.legendoftoys.com`, on the floor tablets.	**Production** and **Dispatch**.	OPERATOR SUPERVISOR DISPATCH	Department PIN, then your QR badge.

Departments, not a free-for-all

The scanner opens on a **Department Select** screen with four tiles: **Production**, **Store**, **Dispatch** and **Attendance**. Each tile leads only to its own stations. A Production device can no longer be switched to a Store or Dispatch station by mistake, and the reverse. This walling-off is deliberate: it removes the wrong-station mis-scans that used to cause real damage, like fresh cars being scanned through a returns station.

1. **Tap your department.** Production or Dispatch. A 6-digit PIN keypad appears on screen (not the phone keyboard, so the PIN is never remembered by the device).
2. **Enter the department PIN.** A wrong PIN just says “try again”; there is no lockout. The unlock then sticks until **1 AM**, even if the app is closed and reopened.
3. **Sign in with your QR badge.** Scan your operator card (or tap **Type ID** and enter your employee ID). This runs once per device per shift and also expires at 1 AM.
4. **Pick a category, then line and station.** Categories show stations in floor order. Choose your line (**L1–L5** for production, **D1/D2** for dispatch) and station, then tap **Launch Scanner**.

> ⚠ **PRODUCTION NOW NEEDS A SIGN-IN TOO**
>
> Every working station, including all Production stations, now asks for your operator QR badge once per device per shift, the same as Dispatch always has. Scan your card after setup. Only the device-free utilities (Lookup and Attendance) skip this.

On the scanner screen

Once launched, the top bar shows your **line and station** (and operator name) on the left and **Logout** on the right. The camera fills the middle; your recent scans run along the bottom. There is **no settings button and no way to switch station in-app**. To change line or station, tap **Logout**, which takes you back to Department Select, then come in again through the guided flow. That friction is intentional.

Shift is automatic

There are no Regular / OT buttons any more. The scanner reads the device clock and tags each scan's shift for you: Packaging counts Regular from 10:00 to 19:00, every other station from 09:00 to 18:00, and anything outside that window as OT. You never set it.

When a scan is rejected

⛔ A WRONG SCAN STOPS YOU WITH A FULL RED SCREEN AND A BUZZ

If a unit is at the wrong stage, belongs to a different run, or the code is the wrong kind, the scanner shows a **full red reject screen with a distinct buzz** and states the exact reason (for example "wrong run, unit belongs to REP-123"). Nothing is recorded. Read the reason, fix it, and rescan. Every station below has been tightened to reject the wrong scan rather than let it through quietly.

Production Stations

OPERATOR SUPERVISOR

The Production department's **Fresh Production** category walks a unit along the line in floor order: Assembly, QC Pass, QC Fail, Workshop, Packaging, then PKG Out. Each station accepts only a unit that is genuinely at the right stage; anything else is rejected with a red screen.

CATEGORY	STATIONS	WHO	YOU SCAN
Fresh Production · lines L1–L5.	Assembly → QC → Workshop → Packaging → PKG Out.	OPERATOR SUPERVISOR	The LOT UPC, or a box label at PKG Out.

Assembly (INW)

Brings a freshly built unit into the system. Scan the unit's `LOT-XXXXXXXX` UPC; a green tick confirms it is inwarded. Assembly checks the unit's product against the line's active run, and that check now applies to **remotes as well as cars**, so a remote from the wrong product is caught the same way a wrong car is. A code that is not a LOT UPC, or a unit already inwarded, is rejected.

QC Pass and QC Fail

The quality check. Scan the unit's UPC only. A unit can be checked only if it has been **inwarded** (fresh) or **repaired** (back from a repair run); anything else is rejected. QC Pass marks it good; QC Fail then asks you to pick the defect codes. On a passing car-plus-remote, the pair is recorded here.

Workshop (WKS)

Where a failed unit is worked on. Scan the unit's UPC. After workshop, a unit goes back through QC.

Packaging (PKG)

Boxes a finished unit. Scan the **car UPC first**, then its remote. Two rules are enforced:

- **QC Pass only.** Only a unit that has passed QC can be packed. A unit at any other stage is rejected.
- **The right pair.** The remote you scan must be *this car's* actual paired remote, not just any remote.

> ⚠ **RE-PACKING IS NOT DONE AT PACKAGING**
>
> Packaging is for fresh units only. To re-pack a unit that has already been boxed or dispatched (for example to swap its channel), use the **Repack** stations, not Packaging. Packaging will reject an already-dispatched unit.

PKG Out

The hand-off from production to dispatch. Scan the box label (`LOT-XXXXXXXX-E/R`). PKG Out accepts exactly two things:

- a **fresh** unit just boxed at Packaging (awaiting dispatch), or
- a **UDR return** that the Store has issued back out for re-dispatch.

A UDR return keeps its original sealed box, so PKG Out re-dispatches it on that same label without re-boxing. Anything that is neither is rejected.

Repair, Outsourced & Repack

OPERATOR SUPERVISOR

Three more Production categories handle units that are not fresh builds: **Repair Run** for units coming back for rework, **Outsourced Run** for finished units returned by a vendor, and **Repack Run** for swapping a packed unit's channel. Each starts at its own first station and then rejoins the normal QC-to-PKG-Out line.

CATEGORIES	FIRST STATIONS	WHO	THEN
Repair Run · Outsourced Run · Repack Run.	Repair · Ext Inwarding · Repack In.	OPERATOR SUPERVISOR	Rejoin QC → Packaging → PKG Out.

Repair Run

A repair run (`REP-NNN`) is a batch of returned or pile units to be reworked. You select the run, then scan each unit in at the **Repair** station to start it. A returned unit issued from the Store's Issue Repair pool is accepted only if it was issued to this exact run. A loose pile unit with no valid label gets a **fresh sticker** applied first, then scanned in. The Repair scan also marks the unit as having been repaired, so anyone downstream can tell a repaired unit from a fresh one by its history. From there the unit goes through Workshop and QC like any other; on a QC Pass the car and remote are re-paired exactly as in fresh production. Parts the repairer needs are requested off the floor as an Ad-Hoc Request linked to the run (see Garage).

Outsourced Run

An outsourced run (`EXT-NNN`) sends raw materials to a vendor who builds and returns finished units. New outsourced runs are **two-phase**: only car, remote and fastener go out; packaging and accessories are issued in-house when the units return. **Ext Inwarding is the finish step**: once the store has issued the finish parts, scan each returned unit in here to sticker it and bring it into the system against the run; it then flows through QC, Packaging and PKG Out like a fresh unit. Ext Inwarding checks that the scanned unit's product matches the run, so a unit cannot be tagged to the wrong job. (Older outsourced runs created before this change keep their original one-step receive.)

Repack Run

A repack run swaps a packed unit from one channel to another. It is two stations:

- **Repack In.** Select the run, then scan the unit's **box label** (`LOT-XXXXXXXX-E/R`). It accepts a box label only, and only a unit that is either just out of **PKG Out** (still on the production side) or one that dispatch has **released at Repack Release**. A dispatch-held unit (Dispatch In or Allocated) that was not released is rejected, and a shipped unit or a bare car / remote is always rejected. Scanning it in releases any allocation and rolls it back so it can be re-boxed for the new channel.

- **Repack Out.** Scan the unit and its destination box. Only a unit actually scanned in at Repack In is accepted, so a fresh unit cannot slip through. The new channel is set automatically from the repack run's request, so you do not pick it by hand.

> ⚠ **DISPATCH UNITS MUST BE RELEASED FIRST**
>
> If the units are with dispatch, a dispatch operator hands them over at **Repack Release** (Dispatch department) before Repack In will take them. Units fresh from PKG Out skip that step. The store also issues the new channel's packaging (box and tray) into the Issue Queue when the run is created.

> ⓘ **SAME HANDLERS, DIFFERENT DOORWAYS**
>
> Repaired and outsourced units flow through the *same* QC, Packaging and PKG Out stations as fresh units; the categories are just separate doorways into one line. So a repaired or vendor-built unit is dispatched exactly like a freshly built one once it passes QC.

Attendance & Lookup

OPERATOR SUPERVISOR DISPATCH

Two scanner tools sit outside the working flow: **Attendance**, for clocking staff in and out, and **Lookup**, for checking any unit's history. Neither moves production along and neither needs an operator sign-in.

TOOLS	YOU SCAN	WHO	RISK
`Attendance` · `Lookup`	An employee QR card, or any unit code.	OPERATOR SUPERVISOR DISPATCH	Both are safe, no sign-in.

Attendance

Attendance is its own tile on the Department Select screen and opens straight into clock-in / clock-out, with **no PIN, no line and no operator gate**. Scan the employee's QR card (or tap **Type ID** and enter it) and confirm to record the event. A 4-digit ID auto-submits; press Enter for a longer legacy ID. Back returns to Department Select.

One scan does the right thing automatically — the system decides clock-in versus clock-out from the person's **shift** and whether they're already clocked in. You never pick "in" or "out":

- **Clock in** — from up to an hour before the shift starts until it ends. Arriving early is fine and is *not* overtime; arriving after the start records how late they were.
- **Clock out** — only **at or after the shift's end time**. Staying late records overtime. You cannot check out early: a scan before the shift ends just shows `Already Clocked In`.
- **Already clocked in** — a second scan mid-shift (or an accidental double-scan) is ignored with that message, so nobody gets clocked out by mistake.
- **See supervisor** — if there's no valid window (e.g. a first scan of the day well before the shift, or after it has ended), the screen says to see a supervisor rather than guess.

> ⓘ **FORGOT TO CLOCK OUT, OR LEFT EARLY?**
>
> No scan is needed for a normal late finish — clock out whenever you leave after the end. If someone forgets entirely, the overnight auto-close credits them to their shift end. For a genuine early departure (half-day, sent home), a supervisor closes the shift or sets a Day status in Redline → Manpower.

Lookup

Lookup is the safe "what is this?" tool, available under the Utilities category in both Production and Dispatch. Scan any code — a LOT UPC, a box label, a remote, or a **part bag** (`BAG-…`) — and it shows that item's details. For a unit: its full history, pairing, box and packing. For a part bag: the part, quantity and product, plus its **pick history** (which run it was picked into, when, and on which device). It changes nothing, so use it freely to check something before acting on it elsewhere.

> ⚒ WHY THESE TWO SKIP THE SIGN-IN
>
> Every working station now asks for an operator badge, but Attendance and Lookup are device-free utilities that record no production work, so they open without one. That also means you can reach Lookup quickly to settle a "where is this unit?" question mid-shift.

PART 5

Repair

Tracking customer repairs and the floor repair queue.

Customer Repairs

`/customer-repairs` SUPERVISOR ADMIN

Customer Repairs is the tracker for warranty and repair cases coming back from customers. Each case (numbered `CR-001`) moves through seven clear stages from “request received” to “dispatched”, so anyone can see where a customer’s unit is and how long it’s been there.

WHAT IT’S FOR	READ OR ACT?	WHO	STAGES
Tracking customer repair cases.	Action: capture, edit, advance.	SUPERVISOR ADMIN	Seven, in order.

The seven stages

1. **Request received**: case logged.
2. **Pickup requested**: courier pickup arranged.
3. **Pickup done**: collected from the customer.
4. **Reached stores**: arrived at the warehouse.
5. **Handed to production**: with the floor for repair.
6. **Repaired · Ready**: fixed, ready to send back.
7. **Dispatched**: sent back to the customer (closed).

The list

Every case with its CR number, customer (name + phone), Order ID, AWB, channel, current stage and **Age**. Age is colour-coded so stale cases stand out: under 3 days grey, 3–6 days amber, 7+ days red. Filter by stage (each chip shows a count) and search by customer, order, AWB or CR number.

Capturing and working a case

1. **+ New Repair.** Only the **customer name** is required; capture fast, fill logistics (order, AWB, pickup) later. New cases start at “Request received”.
2. **Open the case** to edit its details (Save Changes appears once you’ve edited something) and read its event history.
3. **Advance to next stage** with the big yellow button, optionally adding a note. Or use the dropdown to **jump** ahead several stages at once.

ⓘ EVERYTHING IS LOGGED

The case's history records who created it, every stage change (with any note) and which fields were edited, so the full story of a customer's repair is always there. The three repair tiles on the **Returns** screen link straight into this list filtered by stage.

Repair Queue

`/repair-queue` SUPERVISOR ADMIN

The Repair Queue is the pool of physical units on the floor waiting for a repair run: units that failed QC or came back as returns and need rework. Open it to see how many units of each product are waiting so a repair run can be planned.

WHAT IT'S FOR	READ OR ACT?	WHO	RUNS CREATED IN
Seeing units awaiting repair.	Read-only.	SUPERVISOR ADMIN	The Store (Garage) system.

Reading it

A table grouped by product (each product shows a subtotal), then variant rows with Model, Colour, Status and Count, plus a running total at the top. The status tells you why a unit is in the pool:

STATUS	MEANING
QC Fail	Failed the quality check and was pulled for rework.
RTO In	A return that came back in and needs repair.
Scrapped	Marked scrapped during repair.

ⓘ THIS IS A VIEW, NOT A WORKBENCH

The Repair Queue shows what's waiting; the actual repair runs that consume the pool are created in the Store (Garage) system. Don't confuse this with **Customer Repairs**: that tracks customer warranty *cases*, while this is the floor's pool of physical units to rework.

PART 6

Reporting

Cross-cutting production and dispatch reports.

Reporting

`/reporting` ADMIN

Reporting is the analytics hub. Where Overview and Lines show *today*, Reporting lets you look across a date range (production versus target, cycle times, defect and quality trends, line throughput) and export the data as CSV for audits and deeper analysis.

WHAT IT'S FOR	READ OR ACT?	WHO	RANGE
Period reports & data exports.	Read; downloads CSVs.	ADMIN / managers.	10 Days · Week · Month · Custom.

The five sections

SECTION	WHAT IT SHOWS
Production	QC pass, first-pass yield, dispatched, vs-target and run counts, with a daily bar chart and a By Product / By Line table.
Cycle Time	Average QC, PKG and RTD cycle times, with median, fastest/slowest and outlier counts. Green up to 30 min, yellow to 60, red beyond.
Defects	Total occurrences, FPY, top defect codes (coloured by severity) and a severity split, By Code or By Product.
Throughput	Per-station takt times and a per-line bottleneck view that flags the slowest station on each line.
Downloads	CSV exports (see below).

Set the range once at the top (10 Days, This Week, This Month, or Custom) and it applies across the sections.

Downloads

Grouped CSV exports: **Production** (QC View, Plan vs Actual, Defects), **Audit & Compliance** (Customer Repairs, Process Deviations, QC Audit Findings, Scan Violations, Unit Restocks, Damage/Scrap Ledger), and **Movement & Issuance** (Direct Issuances, Cycle Counts, Stock Adjustments). Each downloads for the selected range; an empty range tells you there were no rows.

ⓘ **OPENING THE CSVS**

Files are UTF-8 with a BOM, so Excel opens them cleanly. For richer per-module views, the footer points you to **Garage → Reports**. One export (**Stock Adjustments**) needs the finance reporting permission and is otherwise disabled.

PART 7

Admin

UPC generation, operator management and printing.

UPC Generator

`/upc` ADMIN

The UPC Generator mints and prints sheets of barcode stickers for new units: the `LOT-…` codes that follow each car and remote through its whole life. Open it when the floor needs fresh stickers, and use it to track each print batch from generated to received.

WHAT IT'S FOR	READ OR ACT?	WHO	SHEET
Generating & printing UPC stickers.	Action: generate, print, track.	ADMIN	A3, 609 stickers per sheet.

Generating a batch

1. **Pick the product.** Search by code or name. Products with a remote show a `+ REMOTE` tag; if so, choose the **Car** or **Remote** component (Remote appends an "R" to the code).
2. **Set the quantity** (1–10,000) and any notes.
3. ⚡ **Generate & Print.** The batch is created (you'll see its ID, sticker count, the sequence range (`CODE-from → CODE-to`) and the number of A3 sheets), and the print window opens automatically a moment later.

Tracking batches

Recent Batches lists the last 50, each with its status. The lifecycle:

STATUS	MEANING & BUTTON
Generated	Created. Mark **Sent** when it goes to the printer.
Sent to Print	At the printer. Mark **Received** when the printed sheets are back.
Printed	Printed; mark Received once collected.
Received	Sheets in hand: done.

🖨️ **Print** on any row reprints that batch's sheet (an A3 grid of 13 mm QR stickers, each encoding the unit's LOT code).

NUMBERS ARE MINTED ONCE

Each batch draws a fresh, never-reused block of LOT numbers from the shared sequence: the single source of truth for unit numbering. Generate only what you'll actually print and apply; a generated batch consumes those numbers whether or not the stickers are used.

Operators

`/operators` ADMIN

Operators is the floor staff roster: add and edit people, deactivate leavers, and print each operator's attendance QR identity card. It's the master list that attendance, rostering and performance all draw from.

WHAT IT'S FOR	READ OR ACT?	WHO	DEPARTMENTS
Managing operators & their QR cards.	Managers edit; anyone can print QR.	ADMIN (+ all can view/print)	Assembly, QC, Packaging, Admin, Store, Dispatch.

The roster

A table of Name, Employee ID, Department, Type (In House / Contract) and Status. Search by name or ID, filter by department and status (defaults to Active). Inactive operators show dimmed.

Managing people

- **+ Add Operator**: name is required; pick a department and employment type. The Employee ID is assigned by the system once the operator exists.
- **Edit**: update department, status and phone.
- **Deactivate**: for someone who's left; confirm the prompt. They drop out of the active list and pickers but their history is kept.
- 🖨️ **QR**: available to everyone; prints a name-badge card with the operator's attendance QR.

> ⚠ **TWO DIFFERENT QR CARDS**
>
> The QR printed here encodes the **employee ID** and is for the **attendance** scanner (clocking in/out). It is *not* the scanner-station login card used to log into a workstation; that's a separate card. Don't hand someone an attendance card expecting it to log them into a station.

Print Center

`/print` ADMIN

The Print Center is for reprinting a unit's shipping/packing label when the original is lost, damaged or jammed. Look up an already-packed unit by its batch label or car UPC, pick which printer should produce it, and re-queue the label.

WHAT IT'S FOR	READ OR ACT?	WHO	PRINTERS
Reprinting packing/shipping labels.	Action: looks up & re-queues prints.	ADMIN	L1, L2, L3, D1, D2.

Reprinting a label

1. **Choose Single or Bulk.** Single looks up one unit; Bulk takes a list (one label/UPC per line).
2. **Search.** Enter a batch label or car UPC: a bare number is auto-padded to a full `LOT-…`; a value ending in `-E` or `-R` is treated as a batch label.
3. **Pick the printer.** Select a destination (L1–L3, D1, D2) under "Print To". You must choose one before printing.
4. 🖨️ **PRINT** on a found row (or **PRINT ALL**) re-queues the label to that printer.

> ⚠ **"QUEUED" MEANS SENT, NOT PRINTED**
>
> A green `Queued` badge confirms the job was sent to the printer's queue; it doesn't guarantee paper came out. Check the physical printer. Entries that can't be matched collect in a red "Not found" box; PRINT ALL skips rows already queued so you can't double-print by re-clicking.